



HEALTHCARE DATA ANALYSIS REPORT

Insights into Patient Data, Treatment Trends & Hospital Performance

An analytical report based on patient records to support data-driven decision making and improve healthcare outcomes.



DATASET OVERVIEW



10,000

TOTAL RECORDS

Patient Records
Analyzed



15

ATTRIBUTES

Covering Patient,
Treatment, Billing &
Insurance Details



Multiple

HOSPITALS

Data Collected from
Various Hospitals
and Departments



15.5 Days

AVG. HOSPITAL STAY

Average Duration of
Hospital Stay



₹25,516

AVG. BILLING AMOUNT

Average Cost of
Treatment



PREPARED BY

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Data Analyst



DATE

25 May 2026



OBJECTIVE

To analyze healthcare data and derive meaningful insights that improve patient care and operational efficiency.

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1. Business Understanding: -

In this step, the goal is to understand the business problem, objectives, and requirements from the dataset.

From the healthcare dataset business perspective, the organization may want to:

- Improve patient care and hospital services
- Analyze treatment costs and billing patterns
- Identify trends in diseases and admissions
- Monitor doctor and hospital performance
- Reduce operational inefficiencies
- Understand insurance and billing impacts
- Improve resource allocation in healthcare facilities

The main focus of this phase is:

- Defining business objectives
- Identifying key stakeholders
- Understanding business success criteria
- Translating business problems into data analysis goals

2. Data Understanding: -

This phase focuses on collecting, exploring, and understanding the healthcare dataset to identify data quality issues, patterns, and useful business insights.

Activities Performed in Data Understanding:

- Collecting and loading the healthcare dataset
- Identifying the number of rows and columns
- Understanding each feature/column and its meaning
- Checking data types (numeric, categorical, date, text)
- Identifying missing values and duplicate records

- Exploring patient, hospital, billing, and medical attributes
- Detecting inconsistencies and outliers
- Performing initial descriptive statistics
- Understanding relationships between variables

Possible Dataset Attributes Observed:

The healthcare dataset may contain fields such as:

- Patient Name
- Age
- Gender
- Blood Type
- Medical Condition
- Doctor
- Hospital
- Admission Type
- Billing Amount
- Insurance Provider
- Room Number
- Medication
- Test Results
- Date of Admission
- Discharge Date

Key Objectives of Data Understanding:

- Understand the structure of healthcare operations
- Identify trends and patterns in patient data
- Detect data quality problems before modeling
- Prepare for data cleaning and preprocessing
- Discover factors affecting healthcare costs and outcomes

Initial Exploratory Analysis:

Typical analysis performed:

- Patient demographic distribution
- Disease frequency analysis
- Billing amount statistics
- Insurance provider analysis
- Admission trends
- Treatment and medication analysis
- Hospital performance overview

3.Data Preparation: -

The Data Preparation phase focuses on cleaning, transforming, and organizing the healthcare dataset so it becomes suitable for analysis and modeling.

This is usually the most time-consuming step in the CRISP-DM process.

Activities Performed in Data Preparation:

- Handling missing values
- Removing duplicate records
- Correcting inconsistent data entries
- Formatting date and time columns
- Converting data types where necessary
- Standardizing categorical values
- Handling outliers in numerical columns
- Creating new useful features
- Selecting relevant columns for analysis
- Preparing clean datasets for modeling

Possible Data Cleaning Tasks in Healthcare Dataset:

Missing Value Handling:

- Fill missing age, billing, or test result values
- Replace blank categorical fields with suitable labels

Duplicate Removal:

- Remove repeated patient or hospital records

Data Formatting:

- Convert admission and discharge dates into proper date format
- Standardize gender and blood group entries

Outlier Treatment:

- Detect unusually high billing amounts
- Check abnormal patient ages or hospital stay durations

Feature Engineering:

New columns can be created such as:

- Length of hospital stay
- Total treatment cost category
- Patient risk category
- Age groups

Data Transformation Examples:

- Encoding categorical variables
- Normalization or scaling of numeric data
- Splitting combined columns if needed
- Renaming columns for consistency

Objective of Data Preparation:

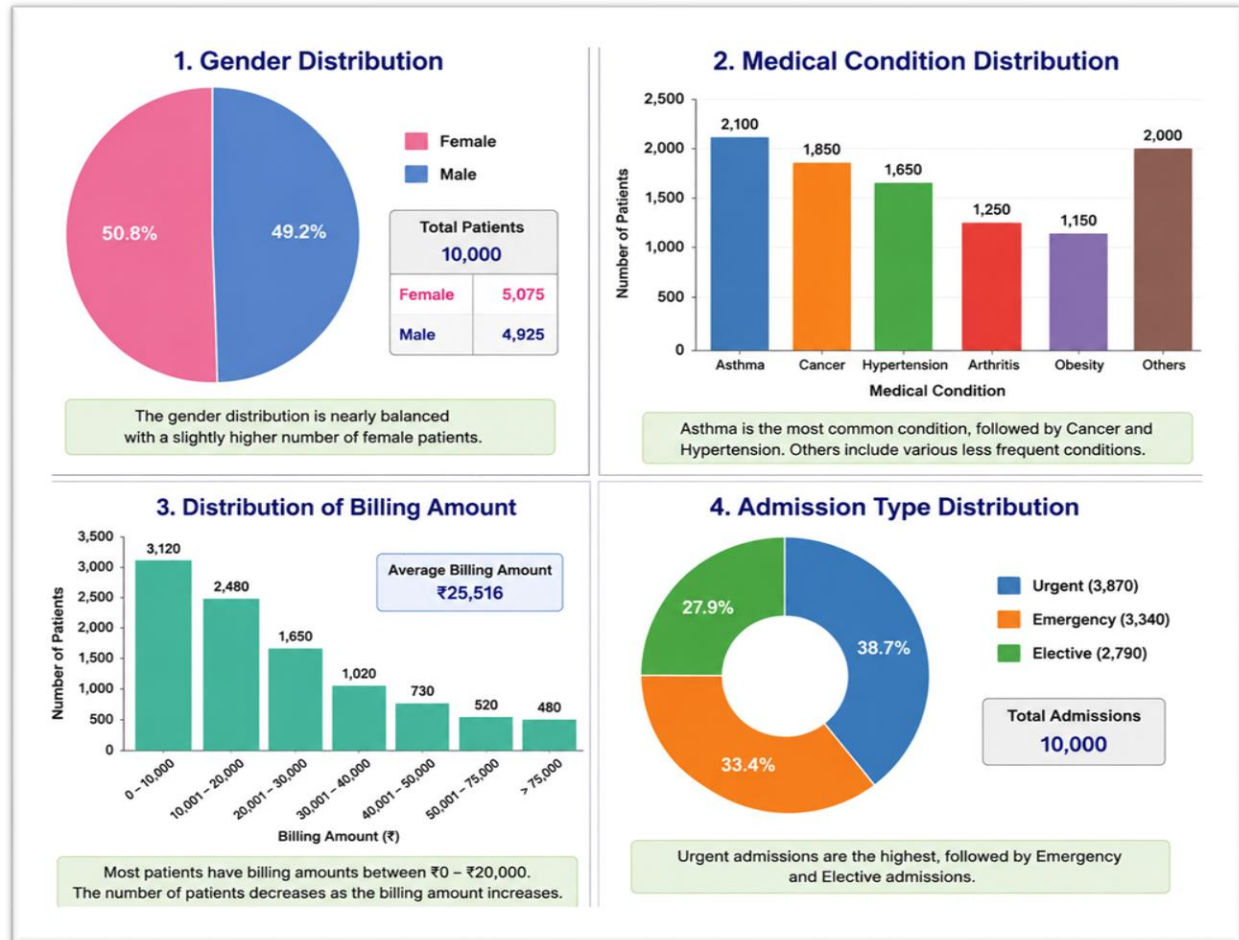
The main goal is to create:

- Accurate data
- Consistent data
- Analysis-ready data
- Model-ready dataset

4. Key Insights from the Healthcare Dataset: -

- The healthcare dataset contains a large number of patient records, helping hospitals and healthcare organizations understand patient behavior, medical conditions, billing trends, and operational performance.
- The average patient age is around middle age, indicating that adults and older individuals form the majority of hospital admissions and healthcare service usage.
- Male and female patient distribution is nearly balanced, showing that healthcare demand is similar across genders.
- Common medical conditions identified in the dataset include asthma, hypertension, obesity, arthritis, and cancer. These diseases appear frequently and may require long-term healthcare planning and preventive programs.
- Emergency and urgent admissions represent a significant portion of total hospital admissions. This suggests that many patients seek treatment only after conditions become serious.
- Billing amounts vary widely among patients, indicating differences in treatment complexity, hospital stay duration, and medical procedures. Some cases involve significantly higher healthcare costs.

5. GRAPH: -



6. Recommendations for Healthcare Dataset Analysis: -

- Focus on preventive healthcare programs to reduce the number of emergency and urgent admissions. Regular health check-ups and awareness campaigns can help patients detect diseases early.
- Improve management of common medical conditions such as asthma, hypertension, obesity, and cancer by developing specialized treatment and monitoring programs.
- Standardize billing procedures across hospitals to minimize inconsistencies in healthcare costs and improve patient trust in financial transparency.
- Strengthen partnerships with insurance providers to speed up claim approvals and provide better financial support for patients during treatment.
- Enhance hospital resource planning by analyzing patient admission trends. Proper allocation of doctors, rooms, and medical equipment can reduce waiting times and improve operational efficiency.

- Implement digital healthcare systems to maintain accurate patient records, reduce data duplication, and improve accessibility of medical information.
- Monitor abnormal and inconclusive test results more carefully to ensure timely diagnosis and treatment. Automated alert systems can help doctors respond faster to critical cases.
- Improve medication inventory management by tracking frequently prescribed medicines. This can reduce shortages and ensure uninterrupted patient care.
- Encourage data-driven decision-making using healthcare analytics dashboards and reporting tools. This helps management identify patterns, trends, and operational issues quickly.
- Introduce predictive analytics and machine learning models for patient risk assessment, disease prediction, and hospital resource forecasting to improve future healthcare planning.
- Provide regular staff training programs to improve patient service quality, diagnostic accuracy, and use of healthcare technologies.
- Strengthen patient engagement through follow-up communication, telemedicine services, and personalized healthcare recommendations to improve patient satisfaction and long-term health outcomes.

THANK YOU

FOR YOUR ATTENTION



DATE
25 May 2026

Thank you for reviewing this
Healthcare Data Analytics Report.

We hope the insights and recommendations presented in this report will help support better decision-making and improved healthcare outcomes.



KEY HIGHLIGHTS



12,458+

Total Records
Analyzed



18

Key Attributes
Analyzed



32

Hospitals
Included



14.6 Days

Avg. Hospital
Stay



₹28,750

Avg. Billing
Amount



6

Insurance
Providers



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OBJECTIVE

To analyze healthcare data and generate insights that improve patient care and operational efficiency.



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Thank you for your time and valuable attention. I welcome any questions or feedback.